Jitu Tesfaye Simessa

Software Engineer

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Professional Experience

Software developer, *Ethiopian Artificial Intelligence Institute* 08/2025 – Present
Addis Ababa, Ethiopia

- Developed and maintained full-stack web applications using React.js, Tailwind CSS, Node.js, and NestJS, delivering responsive and scalable user interfaces with robust backend systems.
- Designed and integrated RESTful APIs to enable secure and efficient communication between frontend and backend services.
- Implemented Role-Based Access Control (RBAC) for dynamic user permissions, improving system security and access management.
- Built reusable UI components and optimized application performance using React Hooks and Redux Toolkit for efficient state management.
- Collaborated with cross-functional teams including designers and backend engineers to deliver production-ready software aligned with business requirements.

Coding Instructor, *Yenneta Codes* 06/2023 – 09/2023
Addis Ababa, Ethiopia

- Taught 30+ students (ages 6-16) fundamental programming concepts and web development during a 3-month internship.
- Delivered personalized code reviews and mentorship to 10 students weekly, fostering an inclusive and engaging learning environment.
- Guided students in writing, debugging 30+ beginner-level programs and small web projects.
- Supported students in static websites, achieving a 100% project completion rate.
- Encouraged teamwork and collaboration through group projects and peer-to-peer coding sessions, strengthening problem-solving and communication skills.

Education

Coding Academy (Backed by Google), *Africa To Silicon Valley* 03/2025 – 01/2026
Relevant Coursework: Data structures and Algorithms
Addis Ababa, Ethiopia

B.Sc. in Software Engineering, *Addis Ababa Science and Technology University* 06/2021 – 06/2025
Addis Ababa, Ethiopia
Relevant Coursework: Internet Programming, Database Systems, Object-Oriented Programming (Java), Operating Systems, Data Structure and Algorithm, Computer Graphics, Embedded Systems, Distributed Systems, Artificial Intelligence and Machine learning.

Projects

Ombudsman Case Management System, *Backend Developer* 05/2026 – Present

- Implemented **address management module** including hierarchical structure for Region, Zone, Woreda, and Branch entities.
- Built and maintained APIs for case handling and submission workflows including submission types, submission modes, and case categories.
- Designed and managed **database relationships** and **Prisma schema** for core system entities.

- Handled data validation, DTO design, and API request/response consistency across modules.
- Collaborated with **two** backend developers to deliver integrated and maintainable system components.

Ethiopian Statistics Service System, *Frontend Developer* 08/2025 – 09/2025

- Delivered the first phase of a role and permission management system, improving workflow efficiency with reusable React components for roles and audit logs.
- Enhanced UI/UX using Tailwind CSS and Material UI, improving usability and maintainability.
- Integrated RESTful APIs with the backend for consistent user workflows.
- Collaborated with backend engineers to align design and functionality.

Skills

Programming Languages
Javascript, Typescript, C++, PHP, Java, Python

Framework and Library
React.js, Express.js, Next.js

Database and Cloud
Mysql, MongoDb Atlas

AI/ML and Tools
Pandas, Numpy, Matplotlib, RoboFlow, Apache Superset, Github, Git, Postman

Achievements

Dean's list, *Addis Ababa Science and Technology University (AASTU)* 10/04/2024
Awarded for outstanding academic performance

A2SV Generative AI Semi-Finalist, *Africa to Silicon Valley* 23/05/2023
Selected among Top 30 teams from 500+ participants.